

Segmentation and Classification of Skin Cancer Diseases Based on Deep Learning

Challenges and Future Decisions

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INTRODUCTION

Why Skin Cancer Detection Matters

Overview

Skin cancer is the most prevalent malignancy worldwide. Over **5 million** new cases are diagnosed annually in the US alone. Early-stage detection dramatically increases survival rates — melanoma detected at Stage I has a **98% five-year survival rate**, compared to less than 25% at Stage IV.



Melanoma

Most aggressive; originates in melanocytes



Basal Cell Carcinoma

Most common; slow-growing, rarely metastasizes



Squamous Cell Carcinoma

Second most common; risk of spreading if untreated

Traditional Diagnosis & Its Limitations

Conventional Diagnostic Methods

→ ABCDE Rule

Asymmetry, Border, Color, Diameter, Evolution — a visual heuristic applied by clinicians

→ Dermoscopy

Magnified skin surface examination; accuracy is heavily operator-dependent

→ Biopsy

Gold standard for confirmation; invasive, costly, and time-consuming

Limitations of Existing CAD Systems

Low generalizability across diverse skin tones and lesion morphologies

Handcrafted feature extraction fails on complex, overlapping lesion patterns

High false-positive and false-negative rates in automated screening tools

Inadequate integration with clinical workflows and real-time diagnostics

CHALLENGES

Key Limitations in Deep Learning–Based Detection



Human Error & Subjectivity

Inter-observer variability among dermatologists leads to inconsistent labeling in training datasets, degrading model reliability.



Dataset Bias

Benchmark datasets (e.g., ISIC) are skewed toward lighter skin tones, causing poor generalization for darker Fitzpatrick scale types.



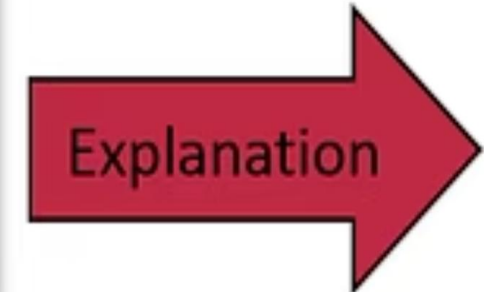
Class Imbalance

Benign lesions vastly outnumber malignant ones, causing models to underperform on the clinically critical minority class.



Black-Box Nature

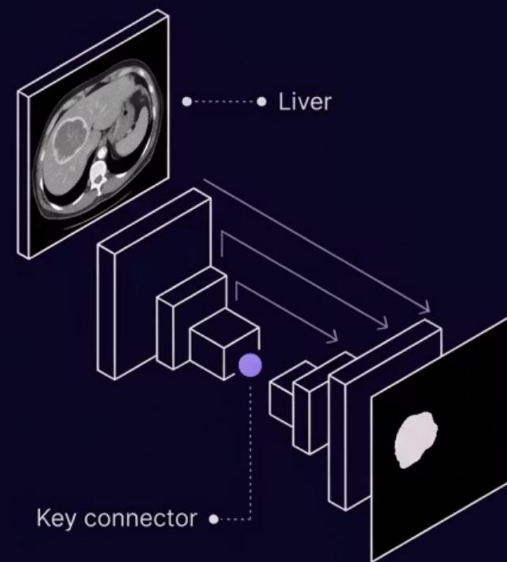
Deep learning models lack interpretability, making it difficult for clinicians to trust or validate automated predictions in practice.



Be

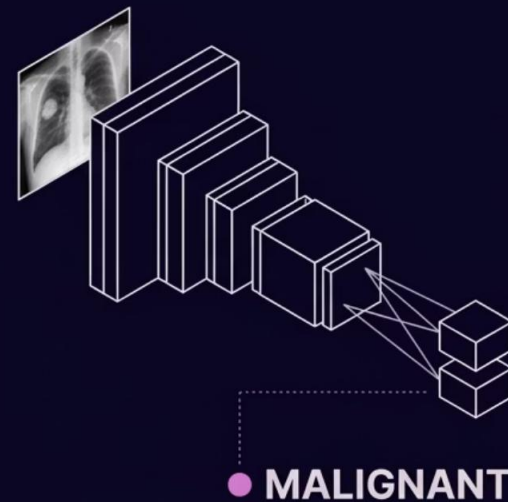
Core Architectures & Techniques

SEGMENTATION



Pixel-wise lesion boundary delineation. U-Net with skip connections.

CLASSIFICATION



Whole-image label prediction. EfficientNet & MobileNetV2.

Why These Architectures?

1 U-Net

Encoder-decoder with skip connections — preserves spatial context for precise lesion boundary delineation

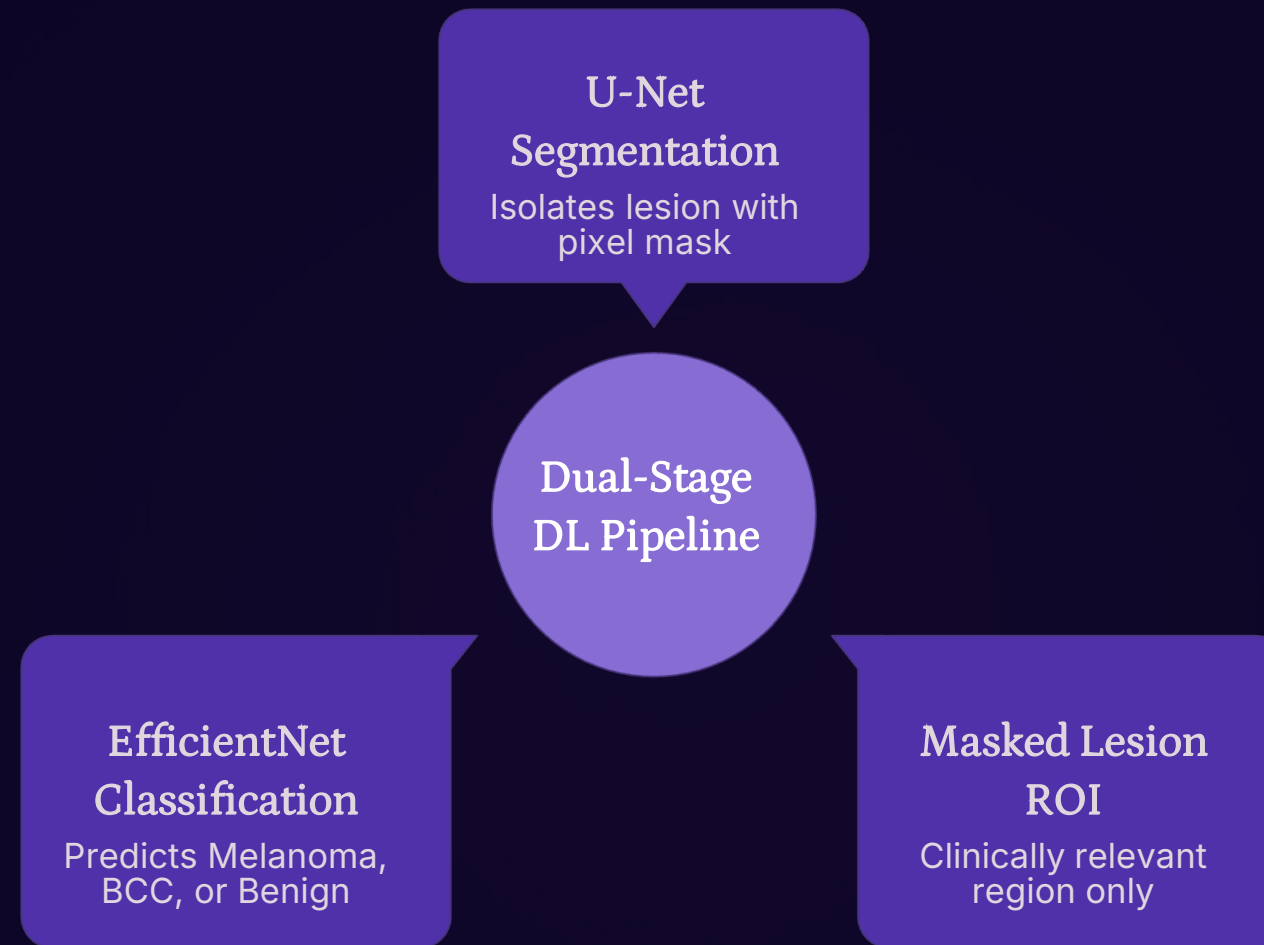
2 EfficientNet / MobileNetV2

Lightweight yet high-accuracy classifiers; ideal for resource-constrained clinical or mobile environments

3 Transfer Learning

Leverages ImageNet pre-trained weights to accelerate convergence on limited medical imaging datasets

Dual-Stage Deep Learning Pipeline



This architecture enables **end-to-end automation** from raw dermoscopic input to structured clinical output delivering faster turnaround, reduced clinician workload, and scalable deployment across healthcare settings.

Improved Accuracy

Segmentation-guided cropping reduces irrelevant background features before classification

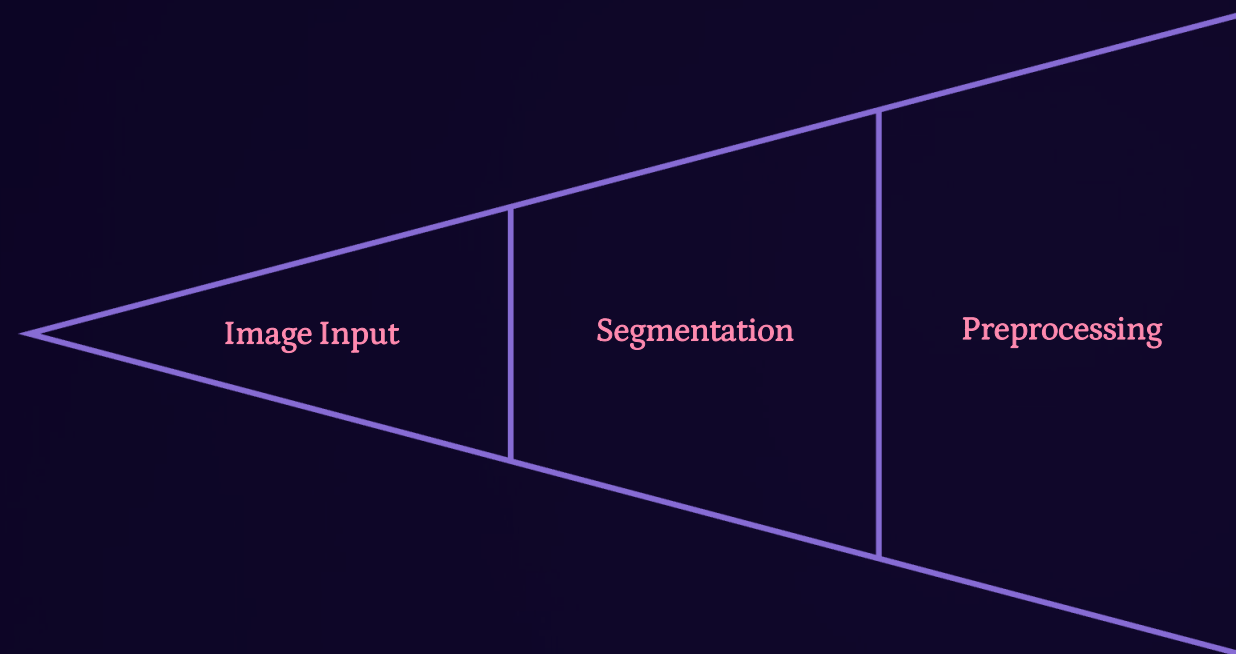
Full Automation

Eliminates manual region-of-interest marking; pipeline runs end-to-end without clinical intervention

Scalable Deployment

Modular architecture allows independent updates to segmentation or classification components

System Architecture & Workflow



Technology Stack



Core ML

Python · TensorFlow / PyTorch · Keras · Scikit-learn



Vision

OpenCV · Albumentations · PIL for preprocessing & augmentation



Frontend

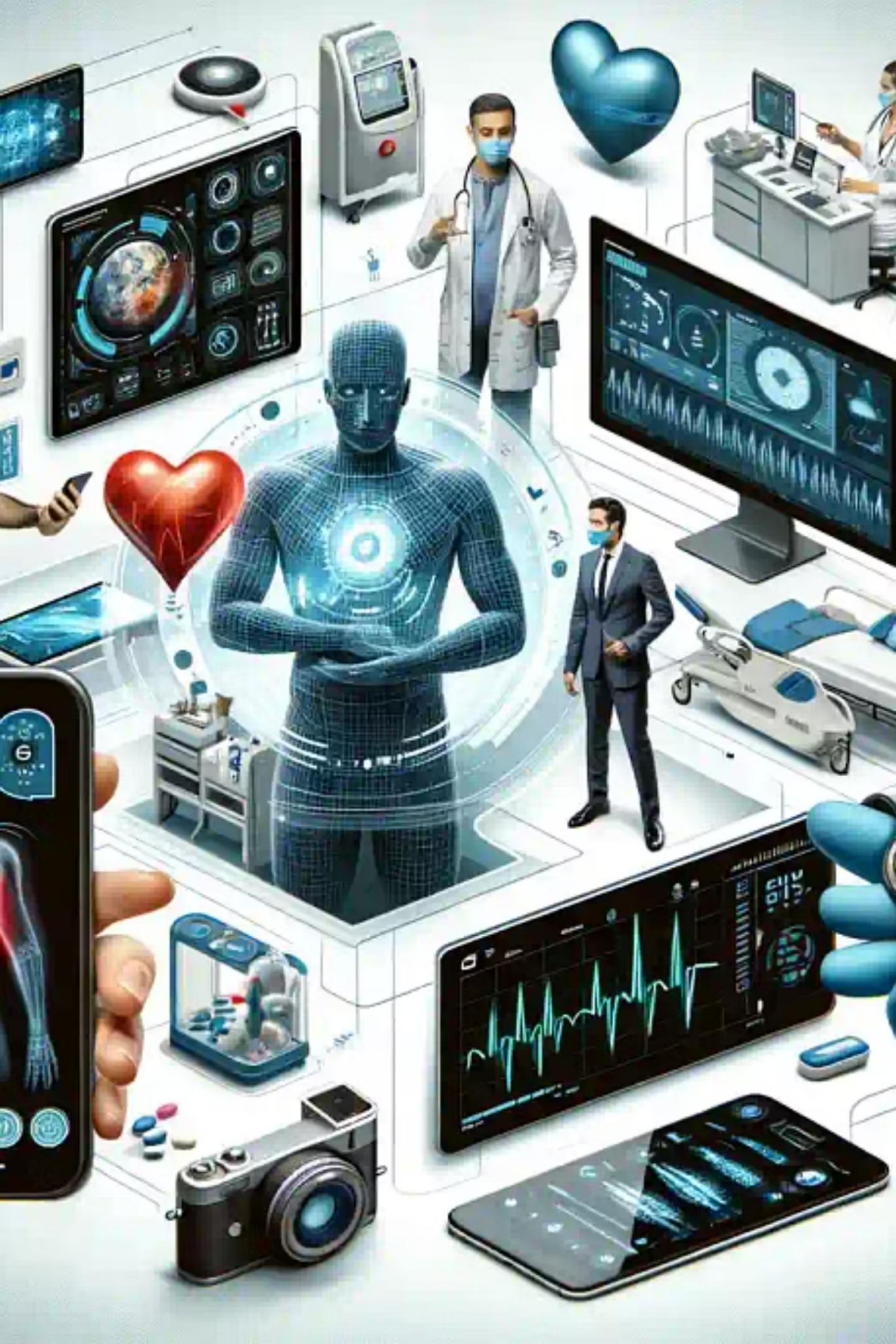
React / Next.js clinical dashboard with REST API backend



Dataset

ISIC Archive — 25,000+ dermoscopic images across 7 lesion classes





RESULTS & FUTURE SCOPE

Outcomes, Conclusions & the Road Ahead

Expected Outputs & Advantages

- **Dual Output**
Segmented lesion mask + malignancy class label with confidence score
- **Speed & Scale**
Sub-second inference enables high-throughput screening at population scale
- **Reduced Human Error**
Consistent, reproducible predictions free from observer fatigue or bias

Future Research Directions

Explainable AI

Integrate Grad-CAM and SHAP to surface model rationale for clinician trust

Mobile Deployment

TensorFlow Lite model compression for on-device inference in low-resource settings

Teledermatology

Cloud-integrated platform enabling remote diagnosis across underserved regions